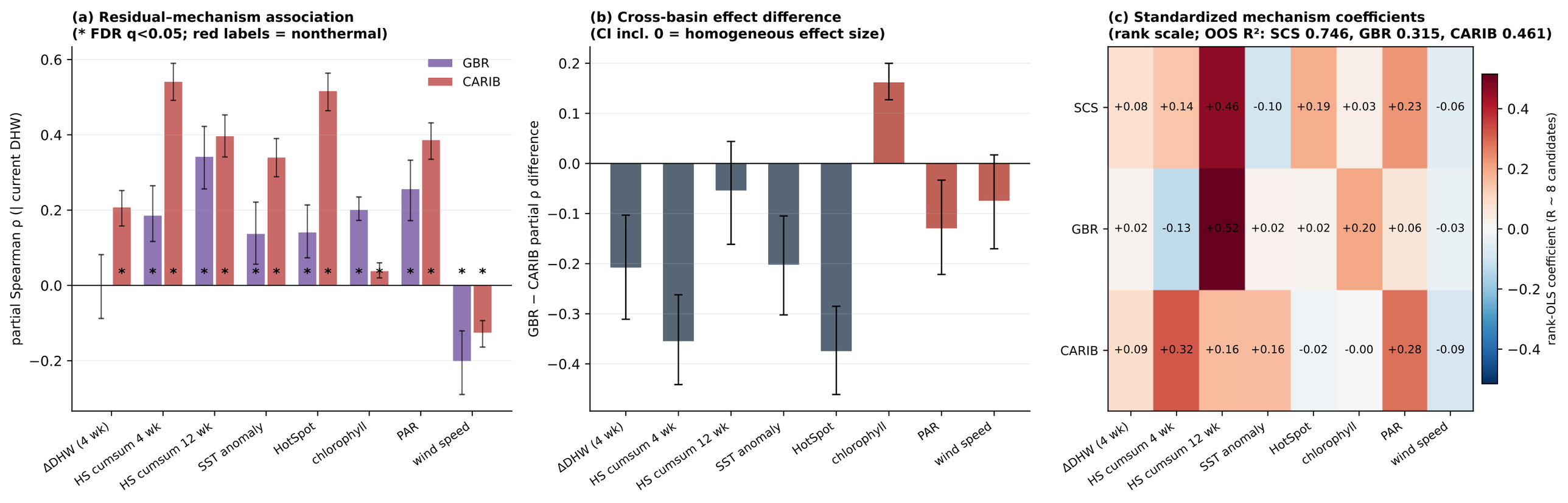
(a) Residual–mechanism association
(* FDR q<0.05; red labels = nonthermal)
partial Spearman ρ (| current DHW)
GBR
CARIB
0.6
0.4
0.2
0.0
−0.2
ΔDHW (4 wk)
HS cumsum 4 wk
HS cumsum 12 wk
SST anomaly
HotSpot
chlorophyll
PAR
wind speed
(b) Cross-basin effect difference
(CI incl. 0 = homogeneous effect size)
GBR − CARIB partial ρ difference
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
ΔDHW (4 wk)
HS cumsum 4 wk
HS cumsum 12 wk
SST anomaly
HotSpot
chlorophyll
PAR
wind speed
(c) Standardized mechanism coefficients
(rank scale; OOS R²: SCS 0.746, GBR 0.315, CARIB 0.461)
SCS
GBR
CARIB
+0.08 +0.14 +0.46 -0.10 +0.19 +0.03 +0.23 -0.06
+0.02 -0.13 +0.52 +0.02 +0.02 +0.20 +0.06 -0.03
+0.09 +0.32 +0.16 +0.16 -0.02 -0.00 +0.28 -0.09
ΔDHW (4 wk)
HS cumsum 4 wk
HS cumsum 12 wk
SST anomaly
HotSpot
chlorophyll
PAR
wind speed
rank-OLS coefficient (R ~ 8 candidates)
0.4
0.2
0.0
−0.2
−0.4